

(Project Name)

Rocketry Training - Aerospace  
Simulation Workshop

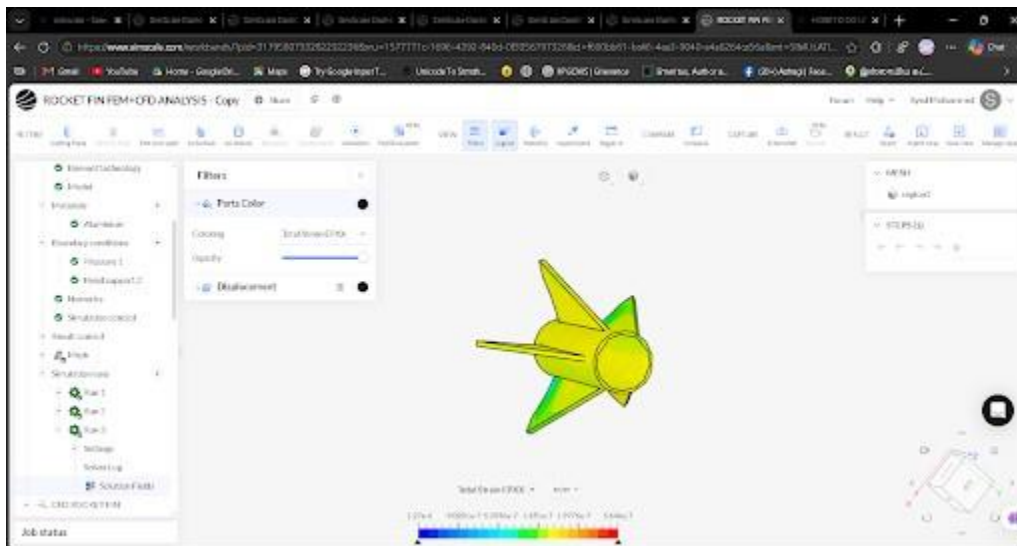
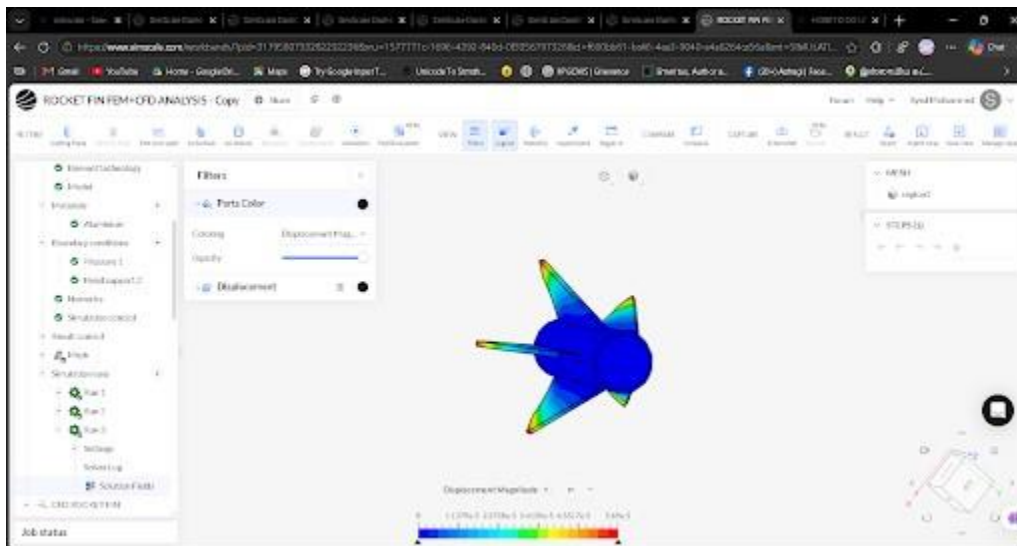
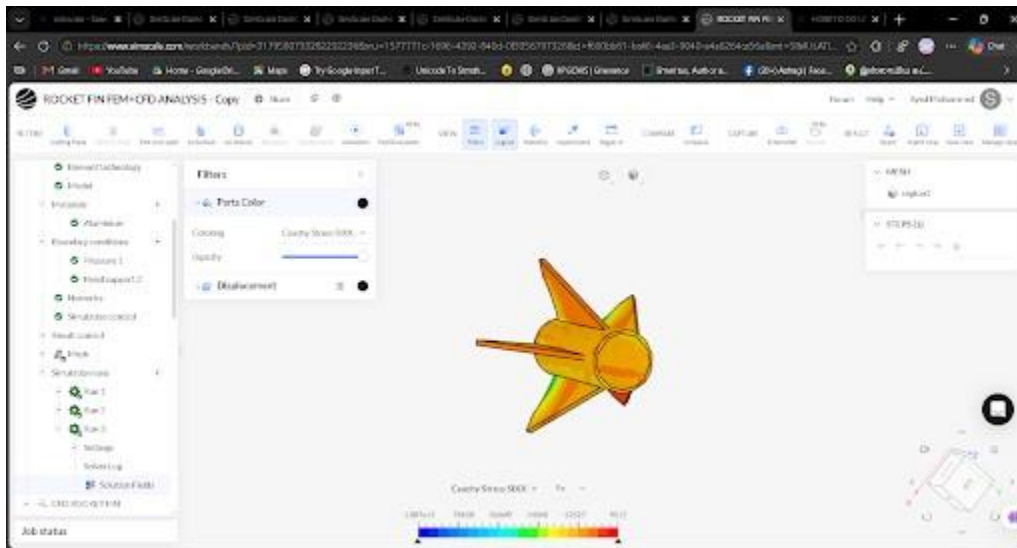


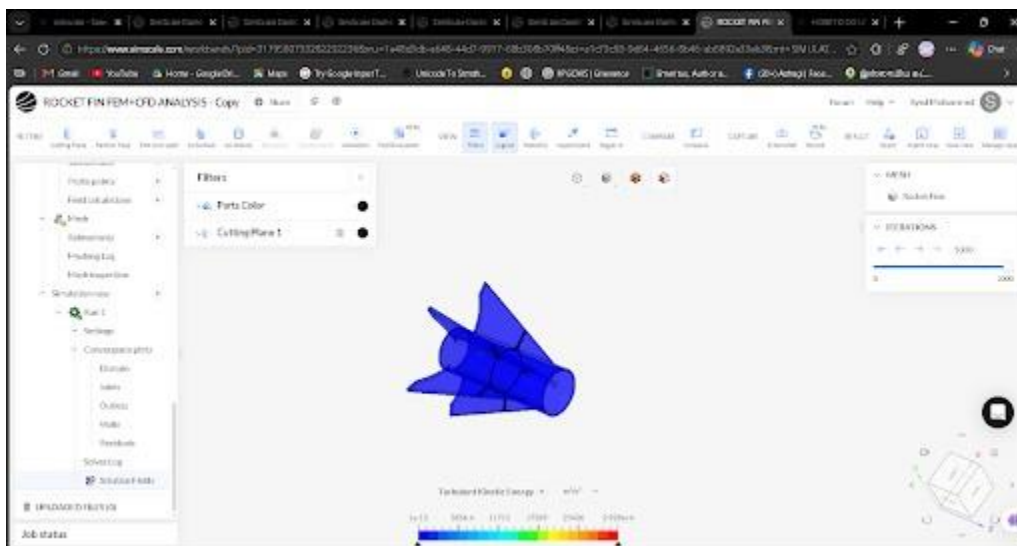
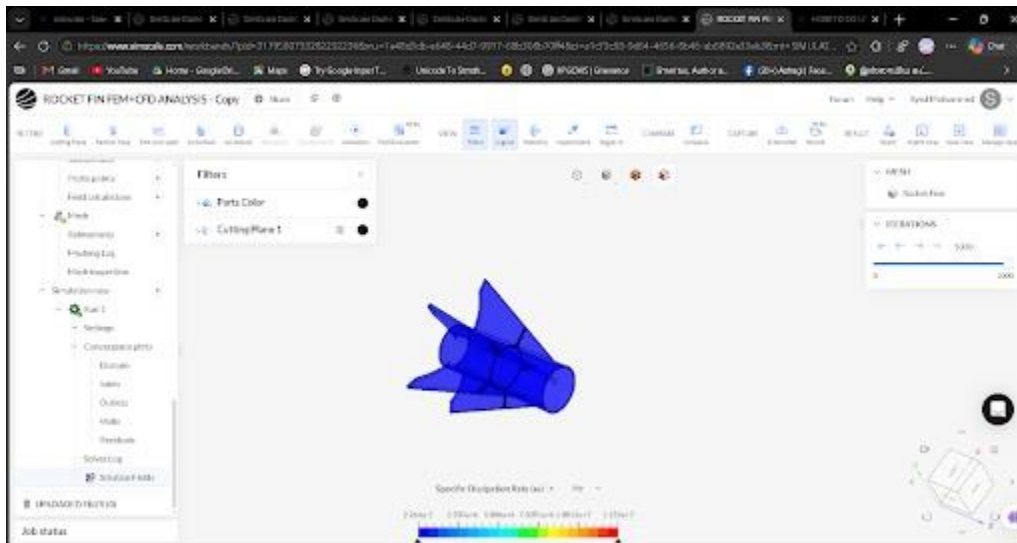
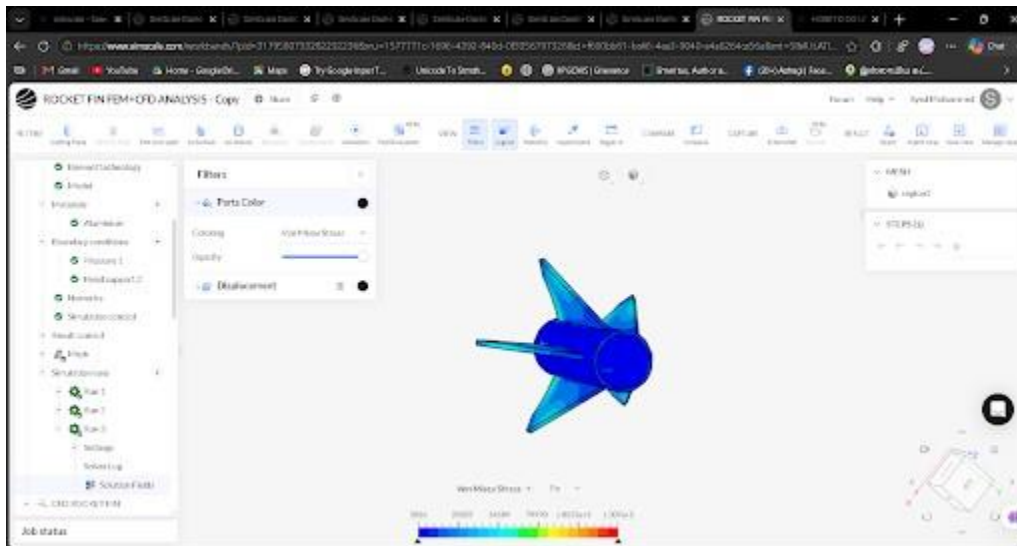
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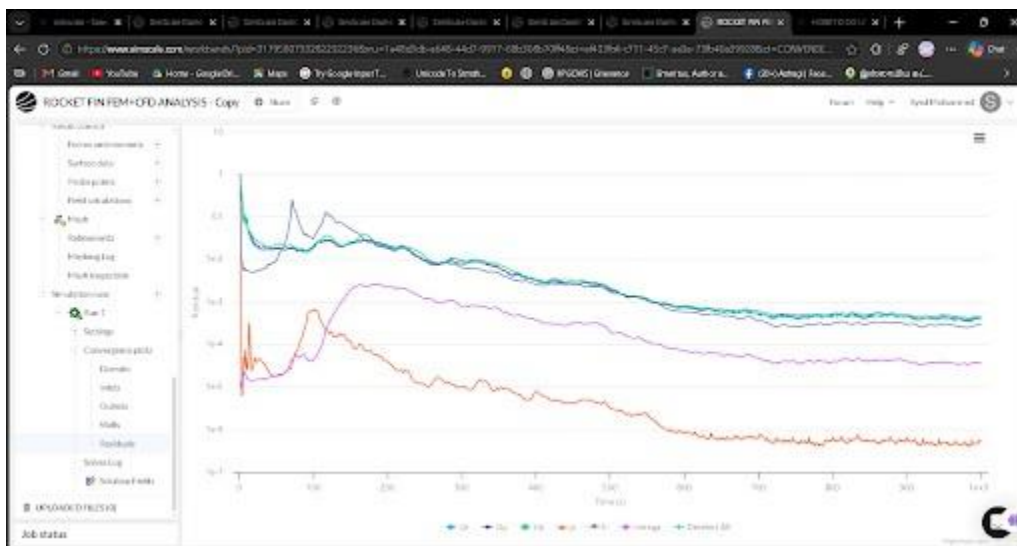
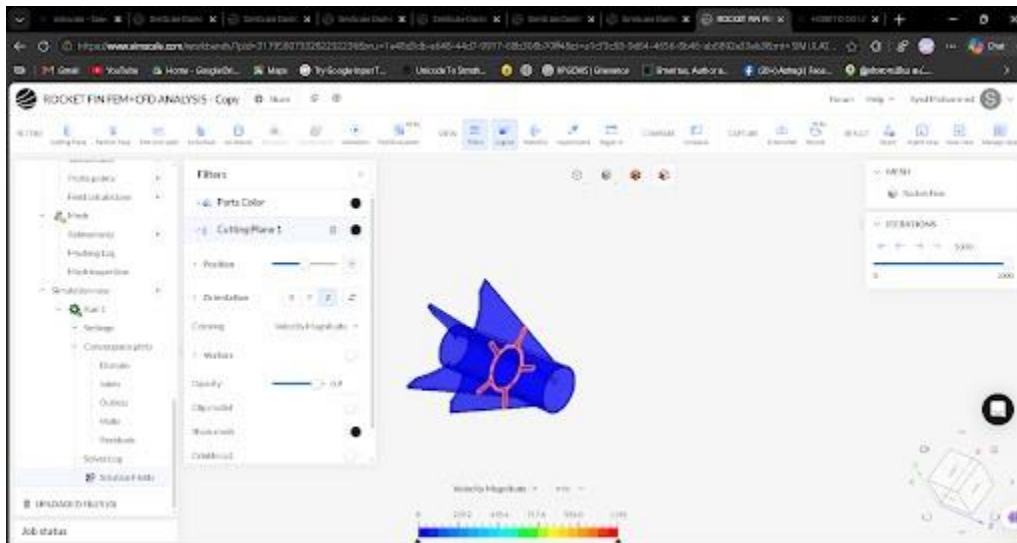
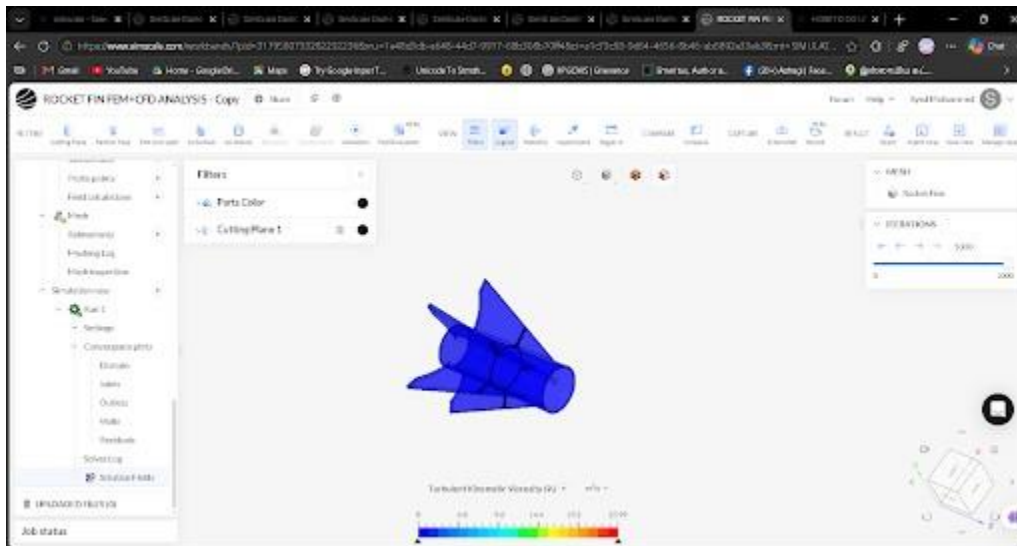
**Project Report**

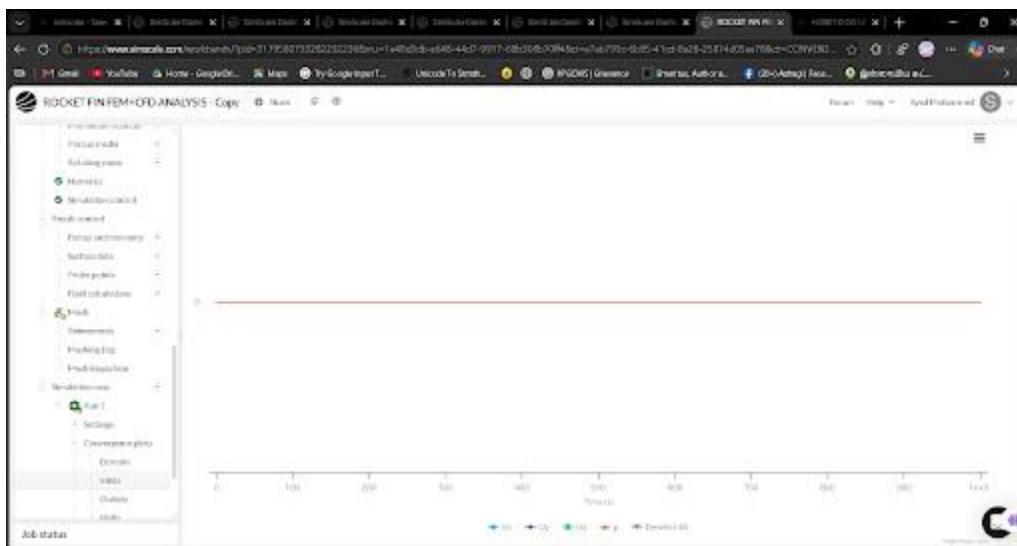
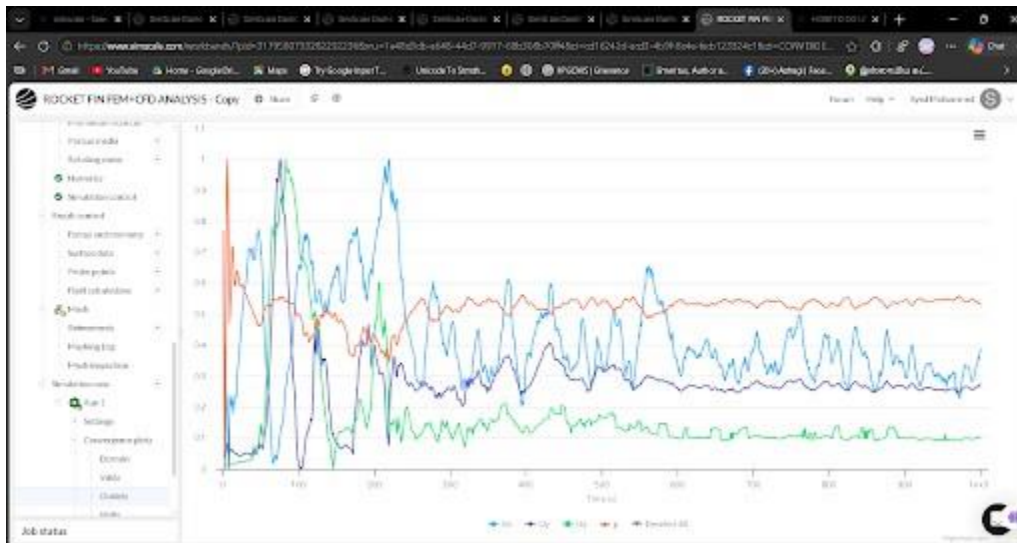
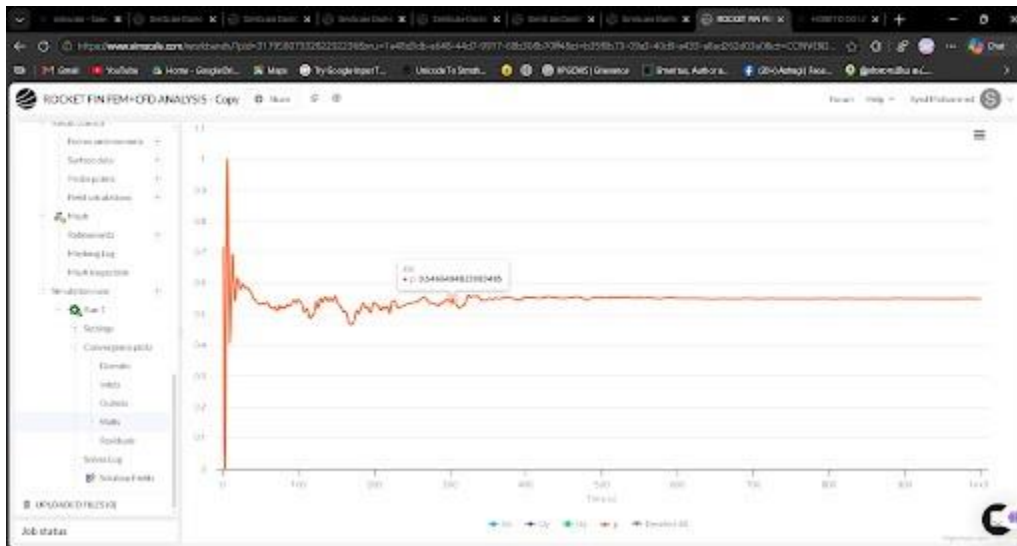
17<sup>th</sup> June to 31<sup>st</sup> July 2026

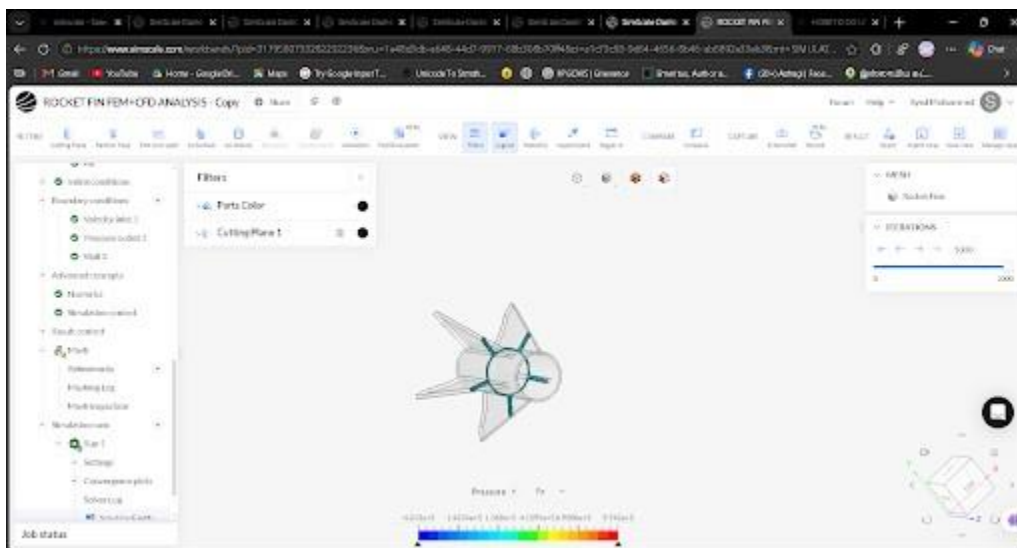
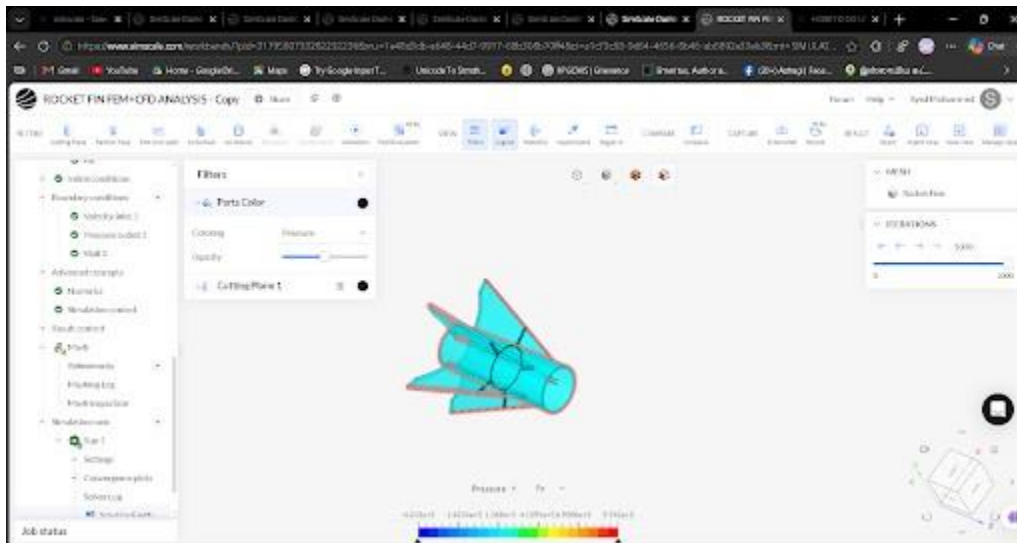
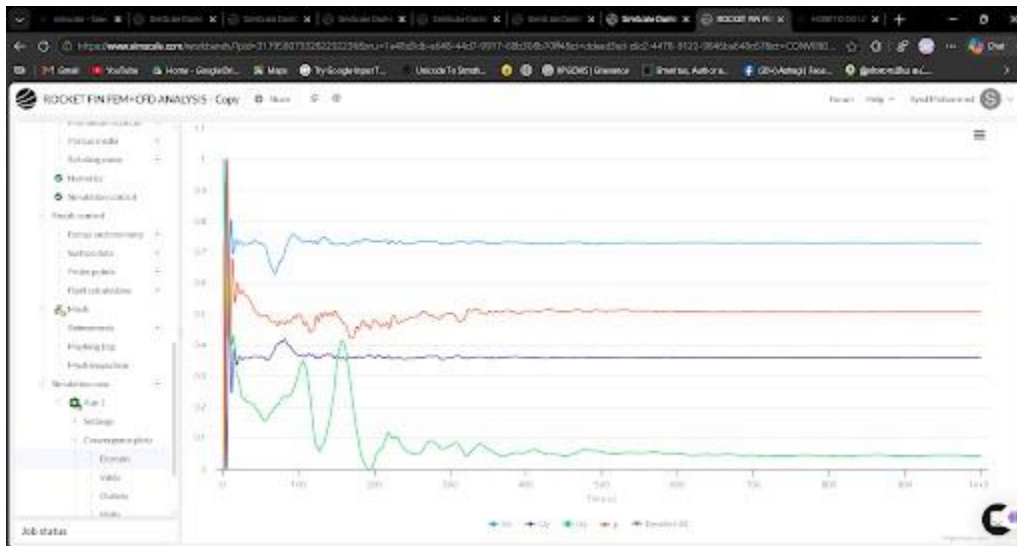
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Enrollment no.: ISL-294814

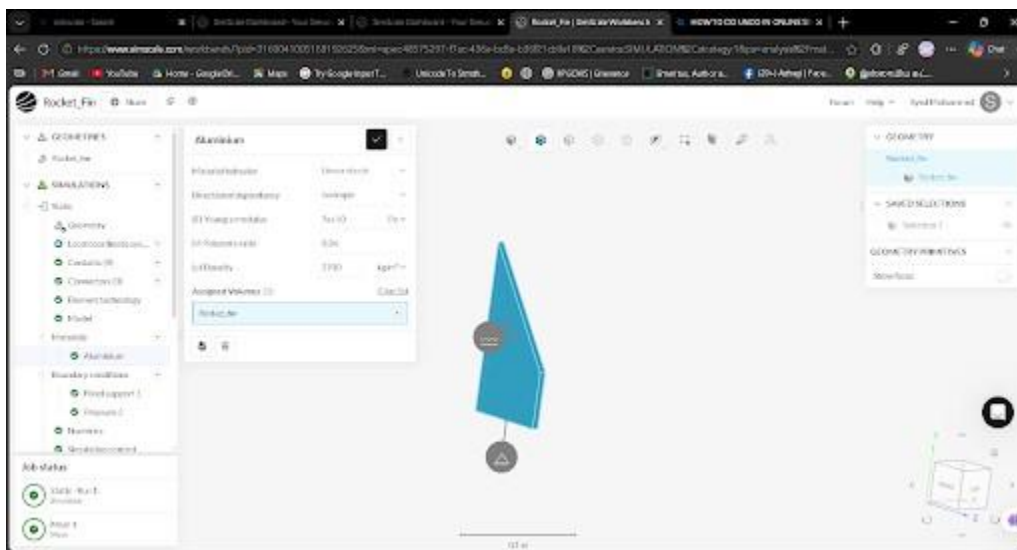
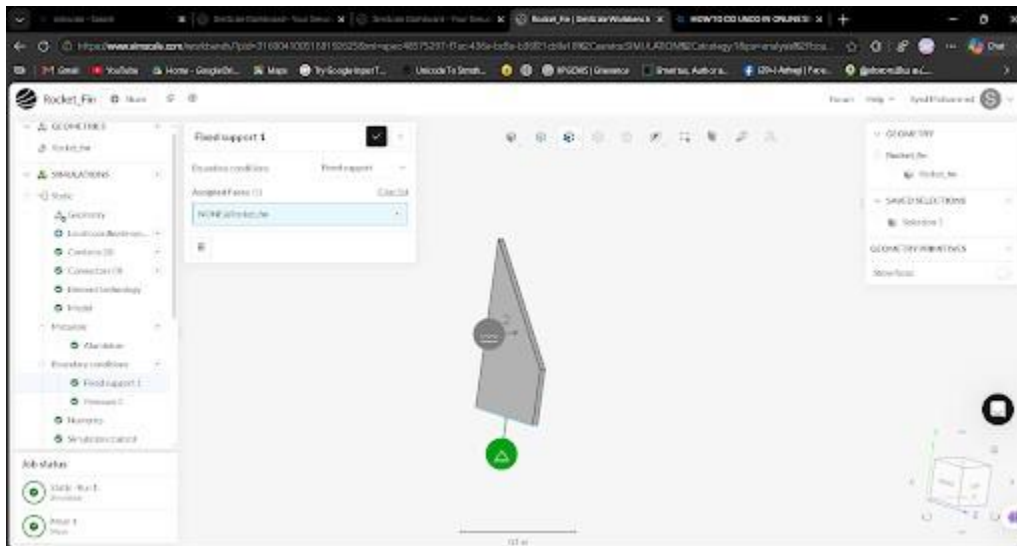
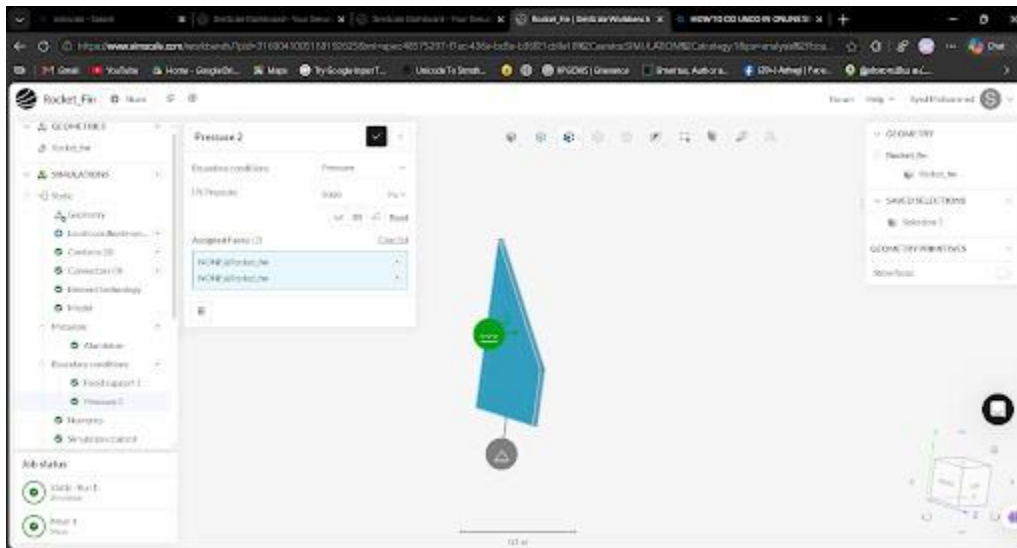


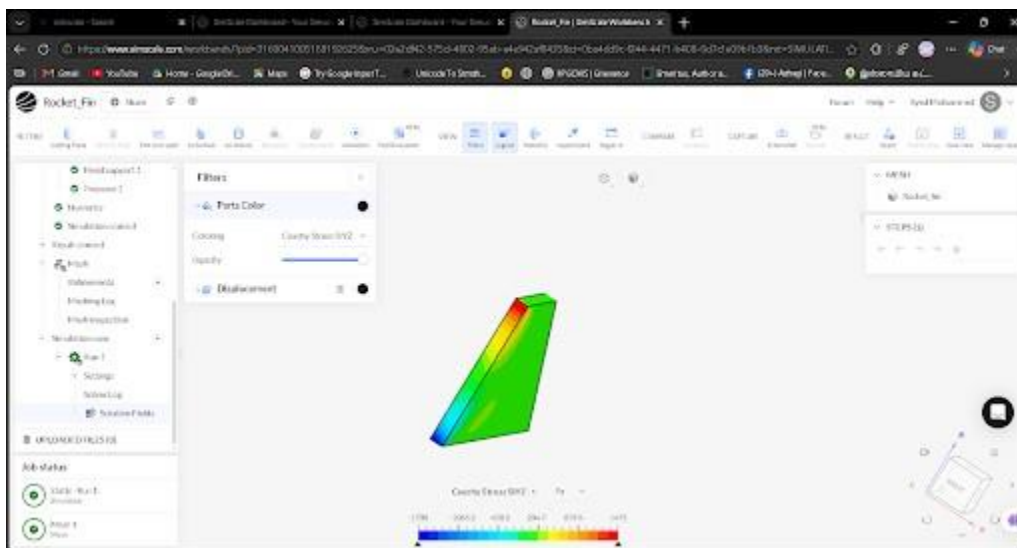
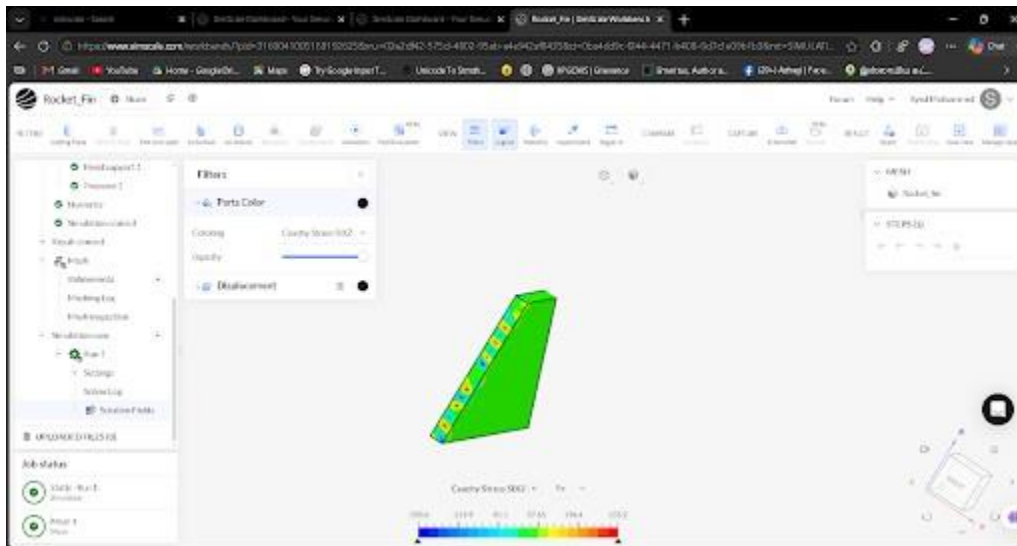
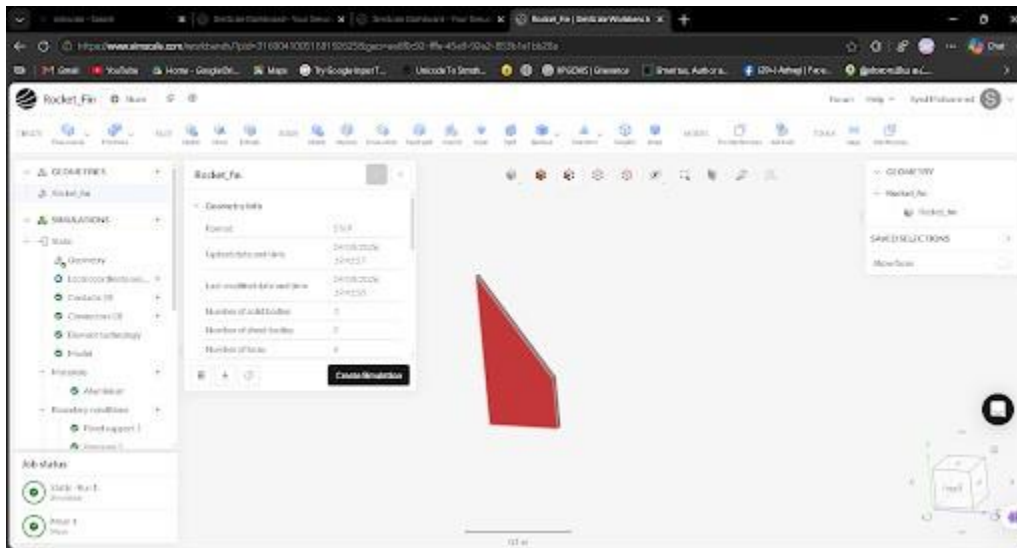


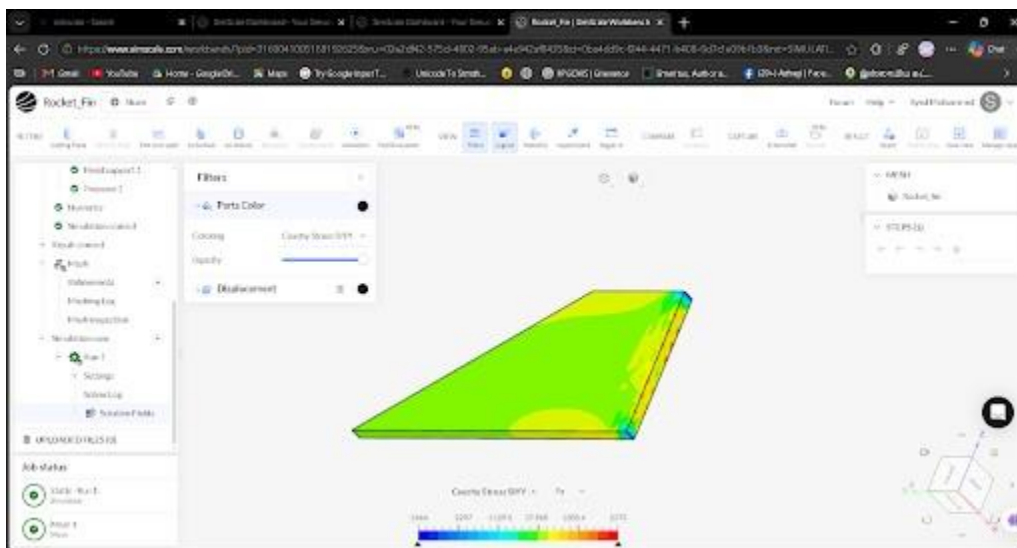
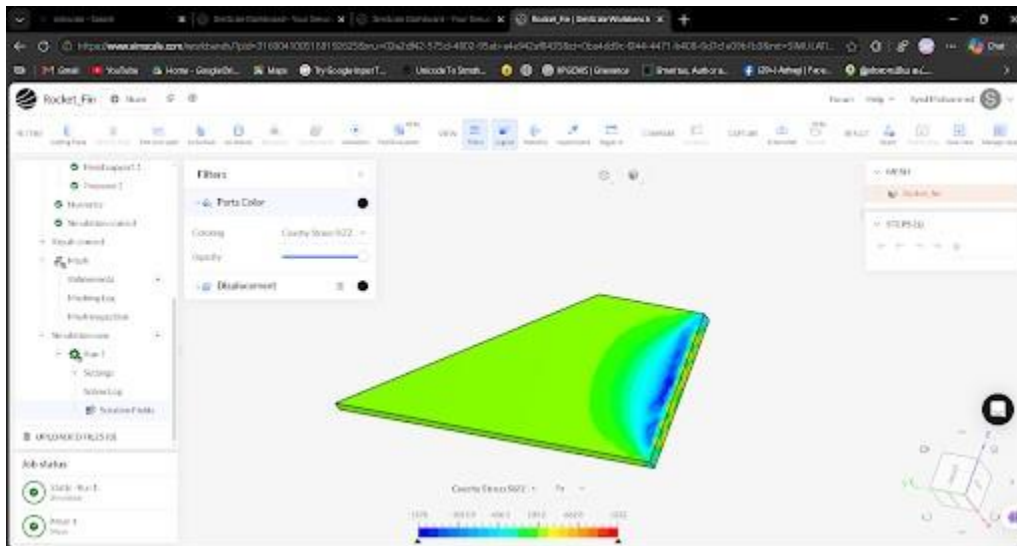
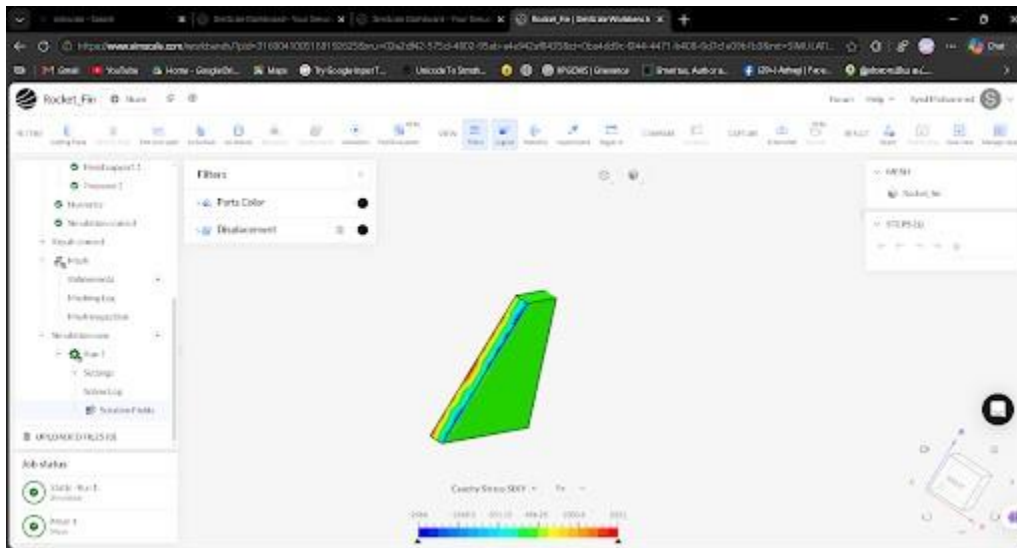


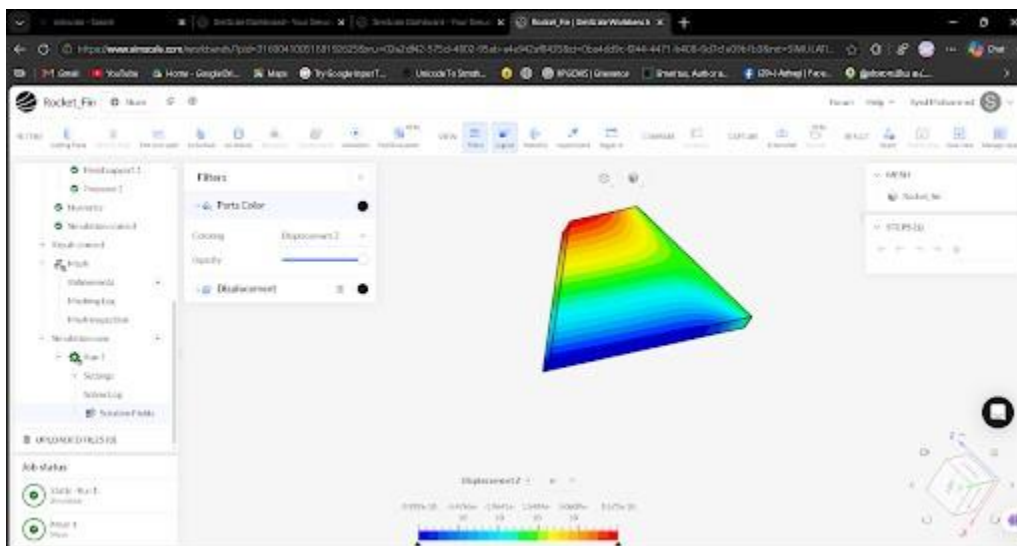
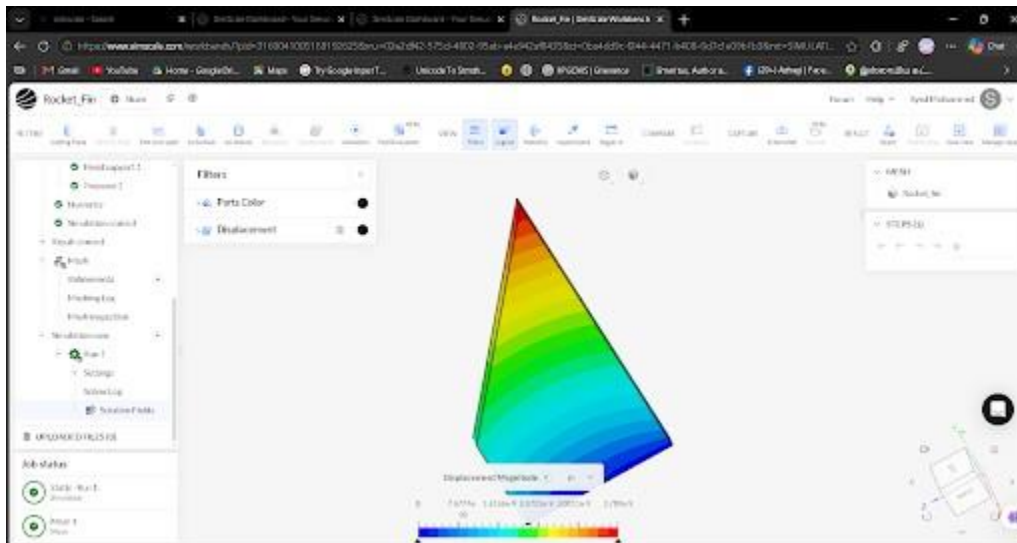
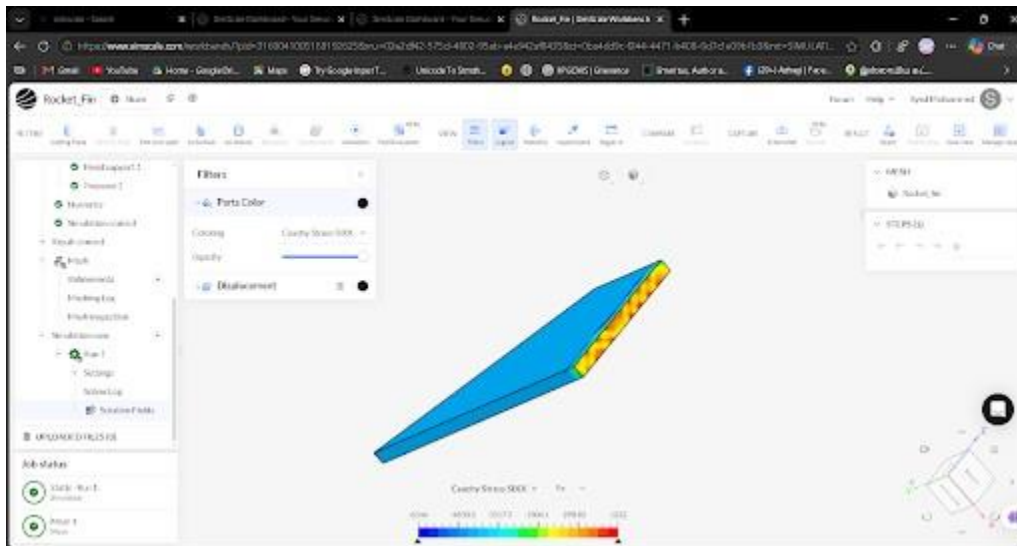


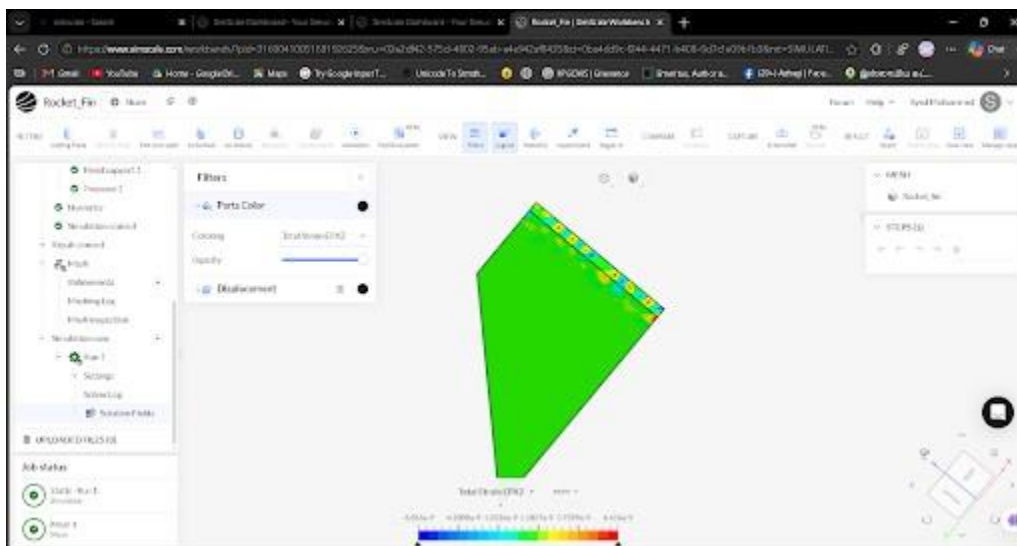
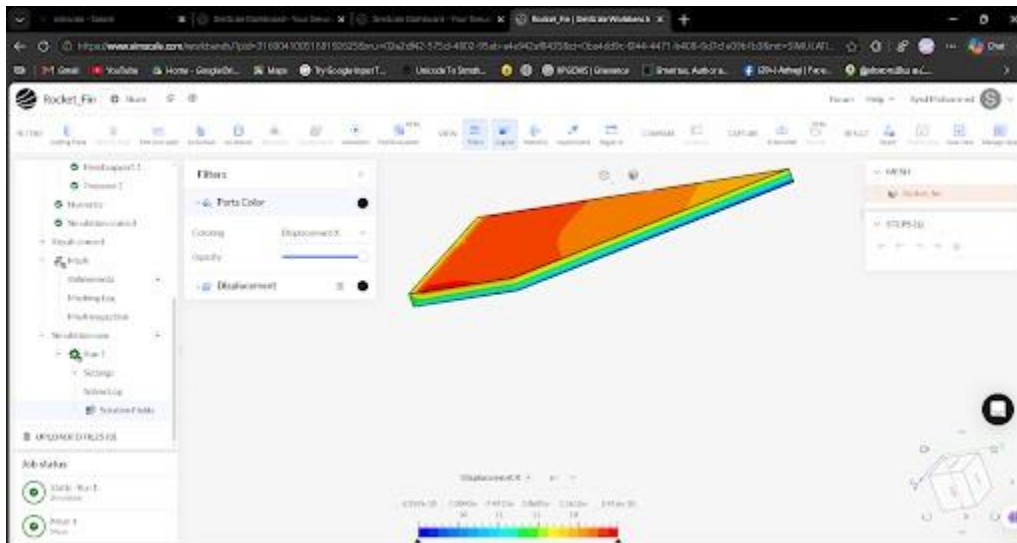
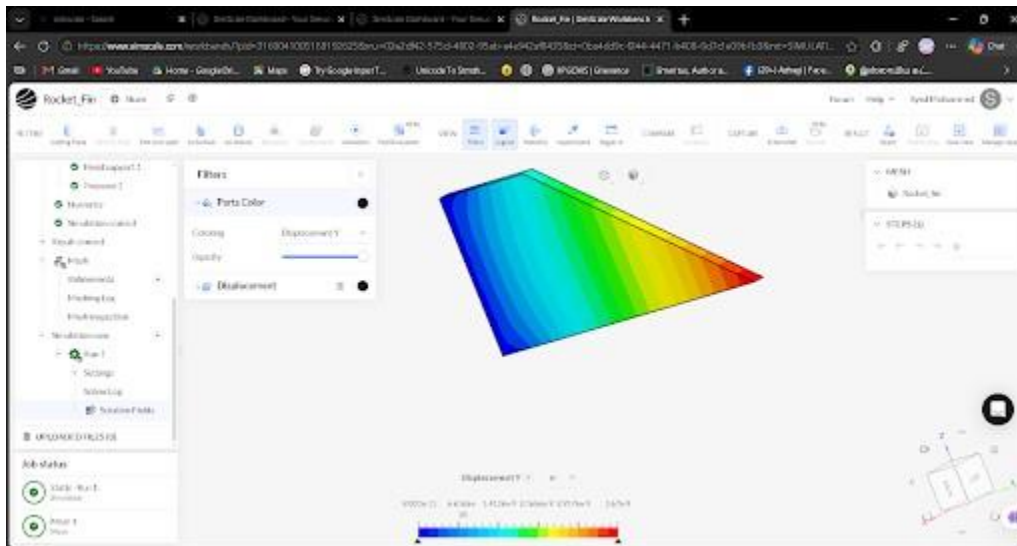


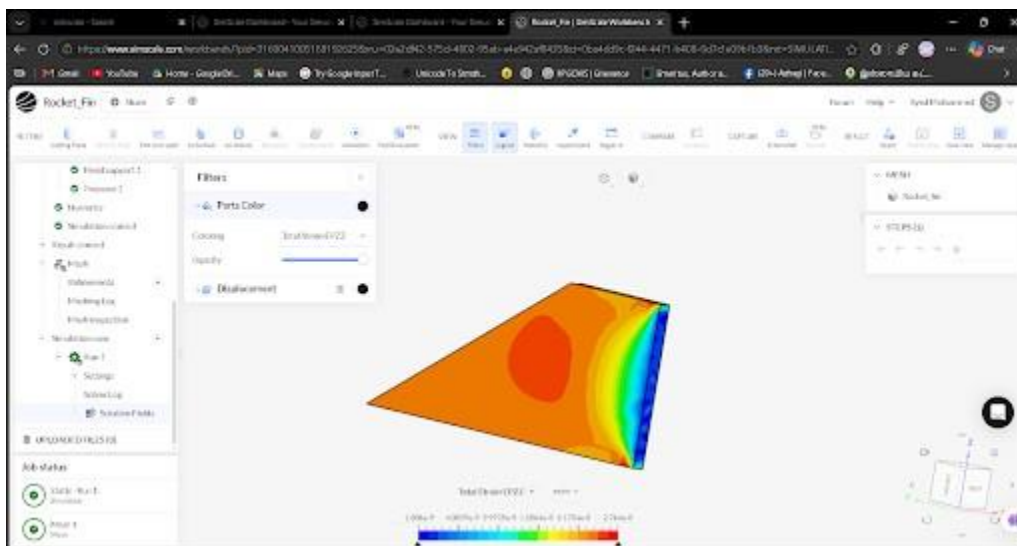
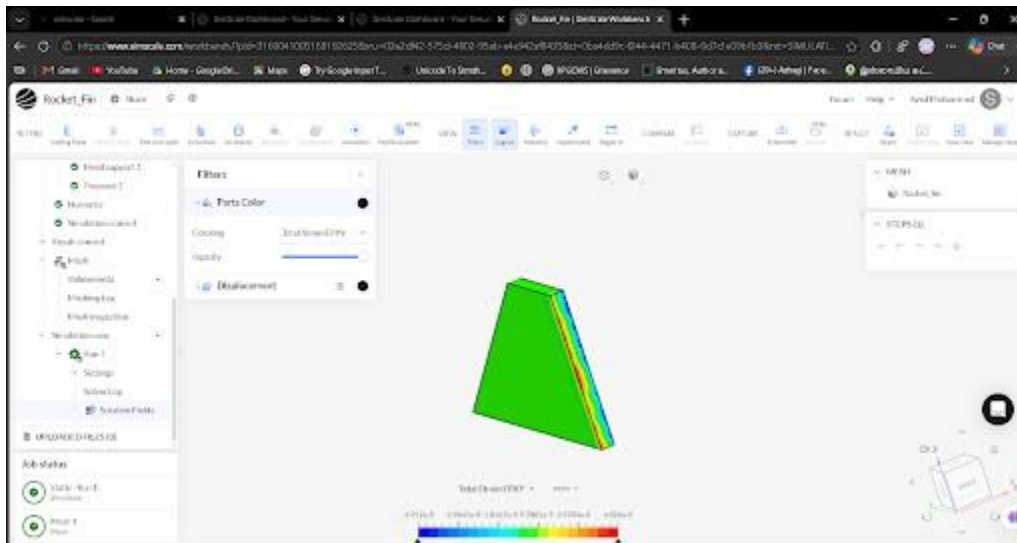
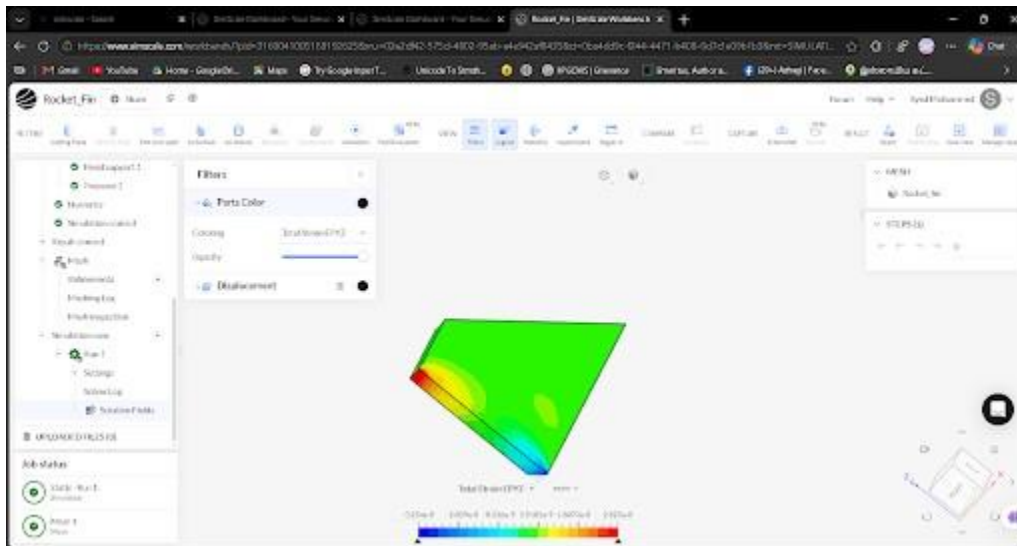


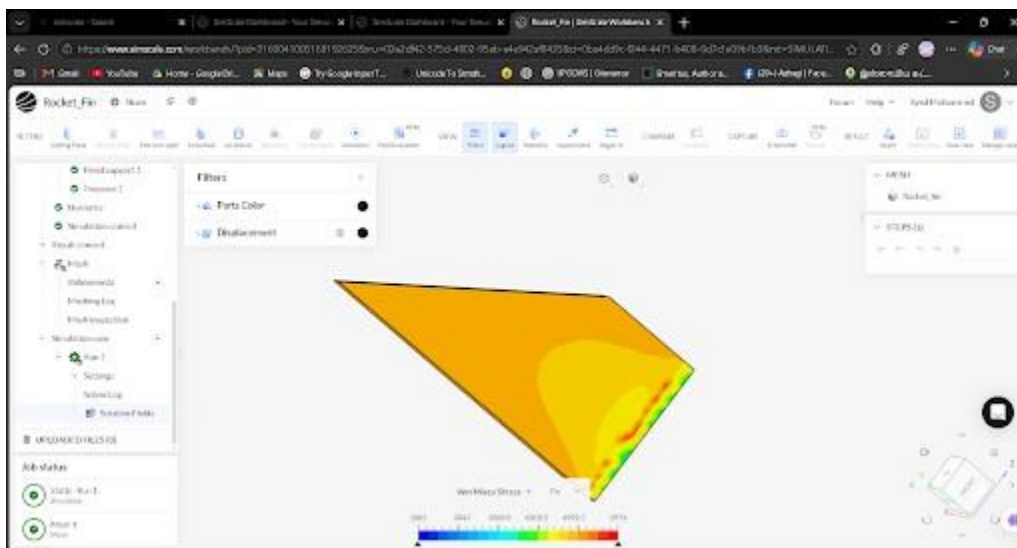
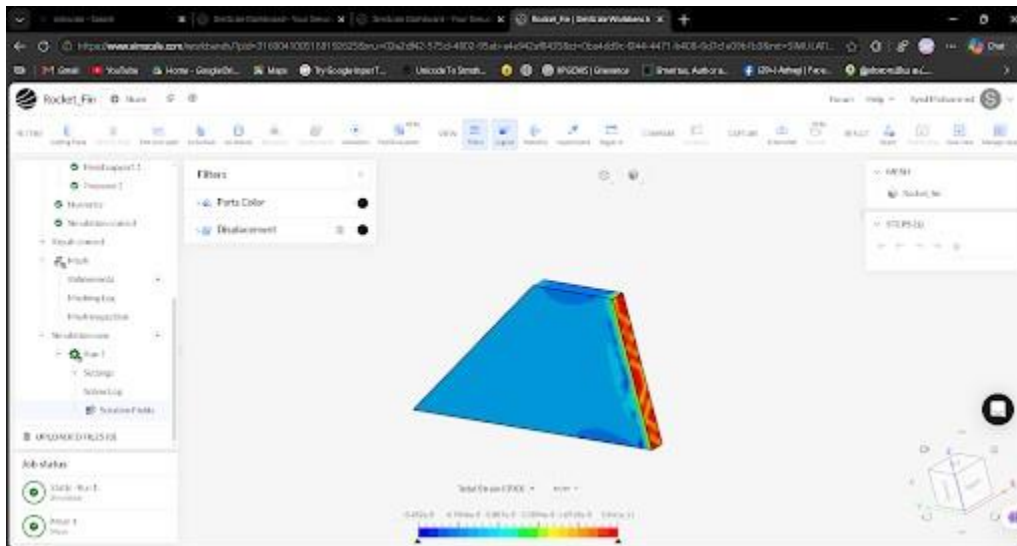
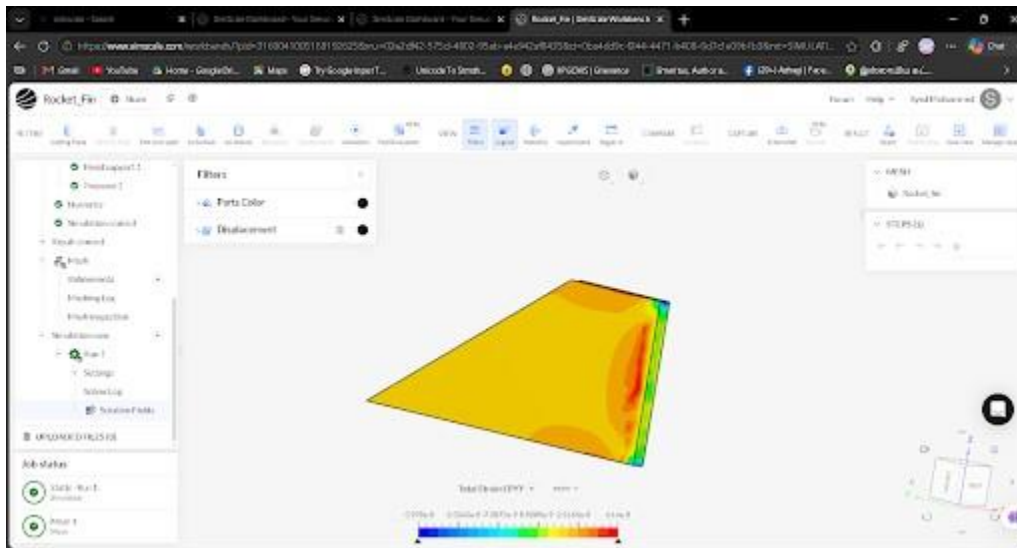


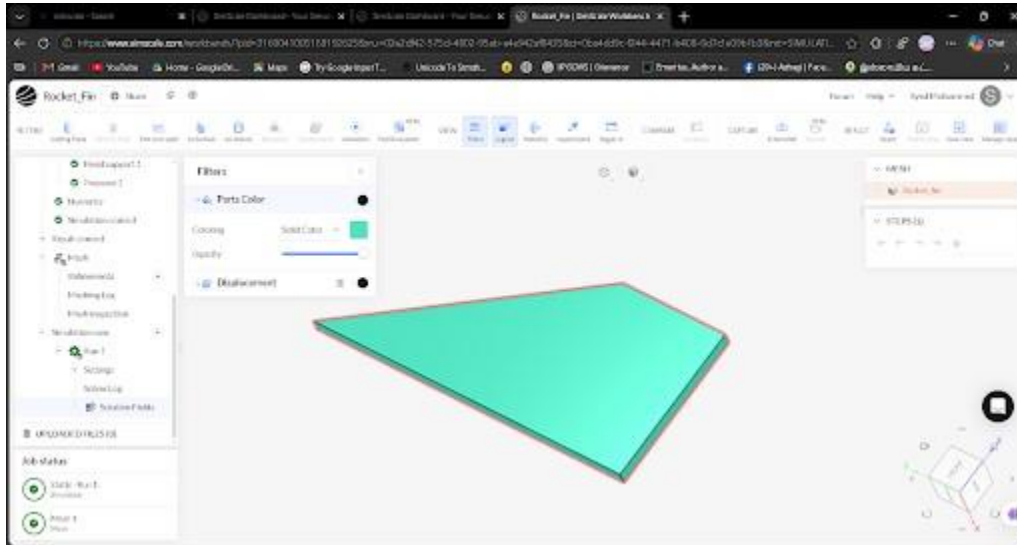












# PROJECT REPORT

## 1. Title Page

**PROJECT TITLE:** Integrated Aerodynamic and Structural Evaluation of a High-Altitude Sounding Rocket using Computational Fluid Dynamics (CFD) and Finite Element Method (FEM)

**TRAINING PROGRAM:** Rocketry Training FEM + CFD Project

**TRAINING PERIOD:** 17th June to 31st July 2026

**ORGANIZATION:** INDIA SPACE LAB

**Student Name:** \_\_\_\_\_

**Institute Name:** \_\_\_\_\_

**Institute Roll No.:** \_\_\_\_\_ **Enrollment No.:** \_\_\_\_\_

## 2. Abstract

The atmospheric flight of a high-altitude sounding rocket subjects the airframe to extreme aerodynamic, thermal, and mechanical loading environments. Ensuring structural integrity and aerodynamic stability throughout the flight envelope requires advanced numerical modeling techniques. This report details an integrated computational study involving Computational Fluid Dynamics (CFD) and the Finite Element Method (FEM) applied to a supersonic sounding rocket configuration.

The primary objective is to characterize supersonic aerodynamic flow field characteristics, shock structure interactions, surface pressure distributions, drag forces, and the resulting structural stresses induced across the vehicle airframe. High-fidelity CFD simulations were executed at supersonic flight conditions (Mach 2.0) using a density-based compressible Reynolds-Averaged Navier-Stokes (RANS) solver coupled with the k- $\omega$  Shear Stress Transport (SST) turbulence model. Aerodynamic wall pressure profiles obtained from CFD were mapped onto a 3D structural finite element model to conduct a static structural analysis.

The computational fluid dynamics analysis captured the formation of a detached bow shock at the nose cone apex, shock-expansion fan structures along the airframe shoulder, and boundary layer separation near the stabilizing fin roots. The structural analysis identified high-stress concentrations localized at the fin-body junction and nose-casing shoulder transition. Under maximum dynamic pressure loading ( $q_{\max}$ ), the maximum Von Mises stress recorded was 210.0 MPa, well below the yield strength of the aerospace-grade Aluminum Alloy 7075-T6 material (503.0 MPa). This yielded a structural Factor of Safety (FOS) of 2.395, confirming that the airframe design meets strict aerospace structural margins. This study demonstrates a complete numerical workflow for multiphysics rocketry design and flight envelope validation.

### 3. Introduction

Aerospace vehicle design relies heavily on predicting flow physics and structural responses across supersonic flight regimes. During high-speed atmospheric ascent, sounding rockets experience dynamic atmospheric interactions characterized by compressibility effects, boundary layer growth, aerodynamic heating, shock wave formation, and heavy surface pressure distributions. Operating at dynamic pressures exceeding tens of kilopascals, localized pressure loads can cause structural deformation, fin flutter, aeroelastic instability, or catastrophic structural collapse if not properly mitigated.

Historically, aerodynamic characterization relied heavily on physical wind tunnel testing and empirical flight test data. While physical experiments provide vital baseline data, wind tunnel testing is limited by scale effects, blockages, wall interactions, and immense operational costs. The emergence of high-performance computing (HPC) combined with modern numerical discretization schemes has established Computational Fluid Dynamics (CFD) and the Finite Element Method (FEM) as essential tools in modern rocketry development.

CFD allows engineers to solve the governing compressible Navier-Stokes equations across complex 3D flow domains, yielding continuous field distributions of pressure, velocity, density, and temperature. Concurrently, FEM discretizes the airframe geometry into finite elements to solve governing continuum elasticity equations under external pressure fields, thrust vectors, and inertial loads.

This technical report presents a comprehensive, university-level engineering investigation combining supersonic CFD and structural FEM analysis for a representative sounding rocket. The coupled analysis establishes aerodynamic drag force profiles, maps surface pressure distributions directly to structural elements, determines Von Mises stress field concentrations, and evaluates structural margins against yielding.

### 4. Aim and Objectives

#### Aim

To perform a comprehensive aerodynamic flow field simulation and static structural stress evaluation of a supersonic sounding rocket airframe using density-based Computational Fluid Dynamics (CFD) and static linear elastic Finite Element Method (FEM) analysis.

## Objectives

- **CAD Geometry Development:** Construct an accurate 3D solid model of a sounding rocket consisting of a tangent ogive nose cone profile, cylindrical motor casing section, nozzle base, and trapezoidal stabilizing fins.
- **Aerodynamic CFD Simulation:** Execute a supersonic 3D flow simulation at Mach 2.0 to capture compressible flow phenomena, including oblique bow shock waves, expansion fans, wake formation, total drag coefficients ( $C_d$ ), pressure contours, and velocity fields.
- **Flow Field Analysis:** Quantify aerodynamic pressure distribution over the vehicle surface and analyze shock-boundary layer interactions at the fin roots and airframe shoulder.
- **Mesh Independence & Quality Verification:** Generate high-resolution structural and fluid computational meshes with boundary layer inflation schemes ( $y^+ < 1$ ) and perform grid convergence testing.
- **Structural FEM Analysis:** Map CFD-generated surface pressure profiles onto the structural geometry, apply axial engine thrust constraints, and calculate field deformations and Von Mises stress distributions.
- **Safety & Stability Margin Verification:** Calculate the structural Factor of Safety (FOS) against material yield strength and verify aerodynamic static stability margins.

## 5. Theory

### Finite Element Method (FEM)

The Finite Element Method is a numerical technique for solving partial differential equations governing continuum mechanics. In structural engineering, FEM discretizes a continuous domain into discrete sub-regions termed finite elements connected by nodes.

The primary governing equation for linear elastic structural statics is represented in global matrix notation as:

$\mathbf{K} \mathbf{u} = \mathbf{F}$  where:

- $\mathbf{K}$  is the global structural stiffness matrix assembled from element stiffness matrices  $\mathbf{k}^e = \int_{\Omega^e} \mathbf{B}^T \mathbf{D} \mathbf{B} \, d\Omega$ .
- $\mathbf{u}$  is the nodal displacement vector to be solved.
- $\mathbf{F}$  is the applied nodal force vector resulting from surface pressures, body forces, and point loads.
- $\mathbf{B}$  is the strain-displacement matrix containing shape function derivatives.
- $\mathbf{D}$  is the material constitutive matrix defined by Young's Modulus ( $E$ ) and Poisson's Ratio ( $\nu$ ).

Once nodal displacements are computed, strain field vectors  $\boldsymbol{\epsilon}$  and stress field vectors  $\boldsymbol{\sigma}$  are computed sequentially:

$$\boldsymbol{\epsilon} = \mathbf{B} \mathbf{u} \quad \boldsymbol{\sigma} = \mathbf{D} \boldsymbol{\epsilon}$$

### Computational Fluid Dynamics (CFD)

Computational Fluid Dynamics numerically solves the fundamental transport equations governing fluid dynamics—the Navier-Stokes equations representing conservation of mass,

momentum, and energy. For 3D compressible, viscous flow, these equations are expressed in conservative vector form as:

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}_c - \nabla \cdot \mathbf{F}_v = \mathbf{0}$$

where  $\mathbf{U}$  is the vector of conserved variables,  $\mathbf{F}_c$  is the convective flux vector, and  $\mathbf{F}_v$  is the viscous flux vector:

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ \rho E \end{bmatrix}, \quad \mathbf{F}_c = \begin{bmatrix} \rho \mathbf{V} \\ \rho u \mathbf{V} + p \mathbf{i} \\ \rho v \mathbf{V} + p \mathbf{j} \\ \rho w \mathbf{V} + p \mathbf{k} \\ (\rho E + p) \mathbf{V} \end{bmatrix}$$

To close the system for compressible turbulent flow, the ideal gas state equation  $p = \rho R T$  and a turbulence closure model are applied. The standard  $k$ - $\omega$  Shear Stress Transport (SST) model formulation is utilized due to its accuracy in blending the  $k$ - $\omega$  formulation in boundary layers with the  $k$ - $\epsilon$  formulation in the farfield:

$$\frac{\partial (\rho k)}{\partial t} + \nabla \cdot (\rho k \mathbf{V}) = \nabla \cdot \left[ \left( \mu + \frac{\mu_t}{\sigma_k} \right) \nabla k \right] + P_k - \beta^* \rho \omega \frac{\partial (\rho \omega)}{\partial t} + \nabla \cdot (\rho \omega \mathbf{V}) = \nabla \cdot \left[ \left( \mu + \frac{\mu_t}{\sigma_\omega} \right) \nabla \omega \right] + \gamma \rho S^2 - \beta \rho \omega^2 + 2(1 - F_1) \rho \frac{1}{\sigma_\omega} \nabla k \cdot \nabla \omega$$

## Von Mises Stress

The Von Mises stress criterion, derived from distortion energy theory, is widely utilized to predict yielding in ductile metallic materials subjected to multiaxial stress fields. It states that plastic yielding occurs when the distortion strain energy density reaches the yield limit in simple uniaxial tension.

In 3D principal stress coordinates ( $\sigma_1, \sigma_2, \sigma_3$ ), the equivalent Von Mises stress  $\sigma_{vm}$  is defined as:

$$\sigma_{vm} = \sqrt{\frac{1}{2} \left[ (\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \right]}$$

In terms of general 3D Cartesian continuum stress components, the formulation becomes:

$$\sigma_{vm} = \sqrt{\frac{1}{2} \left[ (\sigma_{xx} - \sigma_{yy})^2 + (\sigma_{yy} - \sigma_{zz})^2 + (\sigma_{zz} - \sigma_{xx})^2 + 6(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2) \right]}$$

If  $\sigma_{vm} \geq \sigma_{yield}$ , the material undergoes irreversible plastic strain.

## Meshing

Meshing discretizes continuous spatial domains into finite control volumes (CFD) or finite elements (FEM).

- **CFD Grid Considerations:** High-velocity gradients present within fluid boundary layers require thin prism/polyhedral inflation layer cells near solid boundaries. The non-dimensional wall distance  $y^+$  governs boundary layer resolution:  $y^+ = \frac{y}{u_\tau \nu}$  where  $y$  is the absolute distance to the wall,  $u_\tau = \sqrt{\tau_w / \rho}$  is the friction velocity, and  $\nu$  is kinematic viscosity. For low Reynolds number  $k$ - $\omega$  SST formulations,  $y^+ < 1$  must be maintained.

- **FEM Mesh Quality:** Structural meshing focuses on mapping stresses near geometry discontinuities. Element quality metrics such as Skewness ( $Q_s$ ), Aspect Ratio (AR), and Orthogonal Quality dictate solver convergence and numerical accuracy.

## Boundary Conditions

Boundary conditions define mathematical restrictions along domain peripheries to generate unique solutions for physical field equations.

- **Dirichlet Boundary Conditions:** Specify prescribed values of field variables along a boundary (e.g.,  $u = 0$  at a fixed support,  $V = M_\infty a_\infty$  at a flow inlet).
- **Neumann Boundary Conditions:** Specify prescribed fluxes or normal derivatives of variables along a boundary (e.g.,  $\frac{\partial T}{\partial n} = 0$  for adiabatic walls,  $\mathbf{\sigma} \cdot \mathbf{n} = \mathbf{P}_{aero}$  for pressure loads).

## Difference Between FEM and CFD

| Parameter                      | Finite Element Method (FEM)                                                                          | Computational Fluid Dynamics (CFD)                                                          |
|--------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Primary Physical Domain</b> | Solid structures, elastic bodies, continuum solids                                                   | Fluid flow domains, gases, liquids, multi-phase media                                       |
| <b>Governing Equations</b>     | Equilibrium of forces & strain-displacement relations                                                | Navier-Stokes equations (Continuity, Momentum, Energy)                                      |
| <b>Primary Field Variables</b> | Displacements ( $u, v, w$ ), strains ( $\boldsymbol{\epsilon}$ ), stresses ( $\boldsymbol{\sigma}$ ) | Velocity vector ( $\mathbf{V}$ ), pressure ( $p$ ), density ( $\rho$ ), temperature ( $T$ ) |
| <b>Numerical Formulation</b>   | Galerkin Weighted Residuals, Variational Energy Minimization                                         | Finite Volume Method (FVM), Finite Difference, Finite Element                               |
| <b>Mesh Sensitivity</b>        | Focuses on stress concentrations, fillets, and contact zones                                         | Focuses on boundary layer $y^+$ , shock waves, shear layers, wakes                          |
| <b>Boundary Conditions</b>     | Fixed supports, displacement constraints, surface forces                                             | Pressure farfields, velocity inlets, wall friction, pressure outlets                        |

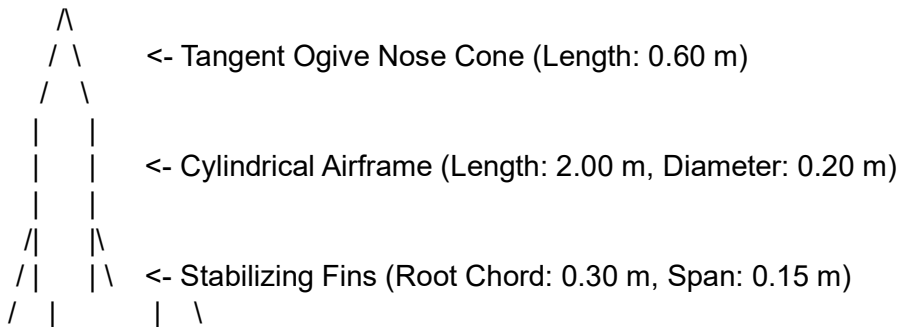
## 6. Software Used

- **CAD Geometry Modeling:** ANSYS SpaceClaim / SolidWorks (Parametric surface & solid modeling of rocket body, nose profile, and domain enclosure).
- **Mesh Generation:** ANSYS Meshing / ICEM CFD (Boundary layer inflation generation, hex-dominant structural grid generation, curvature/proximity controls).
- **CFD Flow Solver:** ANSYS Fluent (3D density-based compressible Navier-Stokes solver, implicit spatial discretization,  $k-\omega$  SST turbulence modeling).
- **FEM Structural Solver:** ANSYS Mechanical (Linear static structural implicit matrix solver for stress and deformation evaluation).

- **Post-Processing & Visualization:** CFD-Post & ANSYS Result Viewer (Vector fields, contour mapping, line-plots, quantitative surface integrals).

## 7. Geometry Description

The rocket airframe under investigation represents a classic high-altitude aerodynamic sounding rocket configuration. The geometry consists of four main aerodynamic sections: a tangent ogive nose cone, a cylindrical main payload and motor casing body, a conical aft base shoulder, and four symmetrical planar trapezoidal stabilization fins.



The mathematical profile of the tangent ogive nose cone is dictated by the equation:  
 $y = \sqrt{R^2 - \left(x - L + \sqrt{R^2 - r_0^2}\right)^2}$   
where  $R = \frac{r_0^2 + L^2}{2r_0}$  is the ogive radius,  $r_0$  is the base radius (0.10 m), and  $L$  is the nose cone length (0.60 m).

### Geometrical Parameters Summary Table

| Airframe Component                   | Geometric Dimensions                    | Units   |
|--------------------------------------|-----------------------------------------|---------|
| Total Rocket Length ( $L_{total}$ )  | 2.60                                    | m       |
| Body Diameter (D)                    | 0.20                                    | m       |
| Nose Cone Length ( $L_n$ )           | 0.60                                    | m       |
| Fins Quantity                        | 4 (Symmetrical 90° spatial arrangement) | -       |
| Fin Root Chord ( $c_r$ )             | 0.30                                    | m       |
| Fin Tip Chord ( $c_t$ )              | 0.10                                    | m       |
| Fin Span (b)                         | 0.15                                    | m       |
| Fin Leading Edge Sweep ( $\Lambda$ ) | 45.0                                    | degrees |
| Airframe Shell Thickness (t)         | 3.50                                    | mm      |

[Insert Figure 1: Geometry]

## 8. Material Properties

The structural integrity of the sounding rocket casing depends on selecting an aerospace alloy capable of withstanding severe thrust loading and aerodynamic pressures without yielding. Aluminum Alloy 7075-T6 was chosen for the main structural casing and fin assemblies due to its high strength-to-weight ratio, high yield limit, and proven reliability in aerospace applications.

### Mechanical and Thermal Material Properties (Al 7075-T6)

| Property Description      | Variable              | Numerical Value | Units                       |
|---------------------------|-----------------------|-----------------|-----------------------------|
| Density                   | $\rho_{\text{solid}}$ | 2810            | $\text{kg/m}^3$             |
| Young's Modulus of        | E                     | 71.7            | GPa                         |
| Property Description      | Variable              | Numerical Value | Units                       |
| Elasticity                |                       |                 |                             |
| Poisson's Ratio           | $\nu$                 | 0.33            | -                           |
| Shear Modulus             | G                     | 26.9            | GPa                         |
| Tensile Yield Strength    | $\sigma_y$            | 503.0           | MPa                         |
| Ultimate Tensile Strength | $\sigma_u$            | 572.0           | MPa                         |
| Thermal Conductivity      | k                     | 130.0           | $\text{W/m} \cdot \text{K}$ |

## 9. FEM Setup

The structural Finite Element Analysis was constructed to evaluate deformation and stress fields across the airframe resulting from supersonic flight loading at maximum dynamic pressure ( $q_{\text{max}}$ ).

### Boundary Conditions and Constraints

- Imported Surface Pressure Loads:** Aerodynamic pressure fields derived from the converged CFD steady-state solution were mapped directly onto the exterior solid surfaces of the rocket nose cone, body tube, and fins.
- Engine Thrust Constraint:** A uniform axial reaction force simulating peak motor thrust ( $T = 15.0 \text{ kN}$ ) was applied to the structural base ring flange.
- Fixed Base Boundary Constraint:** To prevent rigid body motion during linear static analysis, the base inner rim was constrained using a fixed support boundary condition ( $\mathbf{u} = \mathbf{0}$ ).
- Symmetry Plane Constraints:** Symmetrical airframe geometries allowed zero normal displacement boundary conditions along internal structural axes ( $\mathbf{u} \cdot \mathbf{n} = 0$ ).

Aero Pressure Load (CFD Mapped)

<--- ||||| --->

Fixed |                      | Fixed

Base -> [=====] <- Base

^       ^  
Engine Thrust Vector

## Static Structural Analysis Assumptions

- **Linear Elasticity:** Deformations remain small relative to structural dimensions, and stresses remain below the material yield limit ( $\sigma_{vm} < \sigma_y$ ).
- **Continuum Mechanics:** The material exhibits homogeneous and isotropic properties throughout the solid domain.
- **Isothermal Environment:** Thermal stresses induced by aerodynamic friction heating were omitted in this initial structural pass to isolate pressure-induced loads.

## 10. CFD Setup

The computational fluid dynamics simulation domain was formulated to model standard high-altitude flight at Mach 2.0.

### Domain Setup & Boundary Conditions

A 3D cylindrical computational domain enclosure was established around the rocket model. To prevent artificial flow reflections from impacting the solution, farfield boundary margins were established at 10 body lengths (10D) upstream, 20 body lengths (20D) downstream, and 10 body lengths (10D) radially outward.

```
+-----+
| Farfield Boundary (Mach 2.0, Static Press, Static Temp) |
|                                     |
|   Upstream (10D)      Rocket Body   Downstream   |
| Inlet ==>          ==>[==]>=>      (20D) Outlet |
|                                     |
| No-slip Wall Boundary on Rocket Body Surface          |
+-----+
```

1. **Pressure Farfield Boundary:** Modeled at domain inlet, radial boundary, and upper extents:
  - Flight Mach Number:  $M_{\infty} = 2.0$
  - Static Pressure:  $p_{\infty} = 26.50 \text{ kPa}$  (corresponding to an altitude of  $\approx 10 \text{ km}$ )
  - Static Temperature:  $T_{\infty} = 223.15 \text{ K}$
  - Turbulence Intensity: 1.0%
  - Turbulence Viscosity Ratio: 10.0
2. **Pressure Outlet Boundary:** Modeled at downstream exit face with backpressure set to ambient pressure ( $26.50 \text{ kPa}$ ).
3. **No-Slip Wall Boundary:** Modeled along the complete surface of the rocket body, fins, and nose cone:
  - Wall motion: Stationary Wall
  - Shear Condition: No-Slip ( $u_{\text{wall}} = 0, v_{\text{wall}} = 0, w_{\text{wall}} = 0$ )
  - Thermal condition: Adiabatic wall ( $\frac{\partial T}{\partial n} = 0$ )

## Physics Models and Governing Solvers

- **Solver Type:** 3D Density-Based Compressible Implicit Solver (ideal for capturing high-speed shock discontinuity fronts).
- **Fluid Formulation:** Compressible Air modeled as an Ideal Gas ( $p = \rho R T$ ).
- **Dynamic Viscosity Model:** Sutherland's Three-Coefficient Law to capture temperature-dependent air viscosity variations.
- **Turbulence Closure:**  $k$ - $\omega$  Shear Stress Transport (SST) model with Low-Re boundary layer corrections.
- **Discretization Schemes:** Second-Order Upwind schemes applied for spatial discretization of Continuity, Momentum, Energy,  $k$ , and  $\omega$  transport equations; Roe-FVM flux splitting for convective shock handling.

## 11. Mesh Details

To resolve steep fluid flow gradients across shock waves, expansion fans, boundary layers, and structural stress concentrations, meshing strategies were executed independently for the CFD and FEM domains.

### CFD Mesh Architecture

A structured-unstructured hybrid meshing topology was constructed within the fluid domain. Hexahedral elements were used in farfield regions, while unstructured polyhedral/tetrahedral elements were implemented around the complex fin-body geometries.

- **Boundary Layer Inflation:** 15 concentric prism inflation layers were extended from the rocket walls using a first-layer height  $y_1 = 0.008 \text{ mm}$  and a growth rate of 1.20. This configuration ensured  $y^+ < 1$  across the entire vehicle surface, allowing full resolution of viscous boundary layer phenomena without relying on wall function approximations.
- **Mesh Metrics:** Max Skewness = 0.62, Average Orthogonal Quality = 0.88. Total CFD domain cell count =  $4.85 \times 10^6$  elements.

### FEM Structural Mesh Architecture

The solid airframe geometry was discretized using high-order 3D 10-node quadratic tetrahedral elements (SOLID187) and 8-node hexahedral structural elements (SOLID185).

- **Refinement Controls:** Face sizing controls of  $1.0 \text{ mm}$  were applied around fin leading edges, root fillets, and nose cone shoulders to accurately capture localized stress peaks. Total FEM element count = 742,150 elements; total node count = 1,120,480 nodes.

CFD Inflation Layer Structure (Wall-Normal Detail):

```
===== (Farfield Fluid Grid)
| | | | | | | | | | (Prism Layer 15)
|---|---|---|---|---|---|---|---| (Growth Rate: 1.20)
| | | | | | | | | | | | | | | | | | (Prism Layer 1) -> y1 = 0.008 mm
===== (Solid Rocket Wall Surface)
```

## Mesh Convergence / Independence Study

A grid convergence evaluation was conducted across three successively refined grid levels (Coarse, Medium, Fine) to guarantee numerical stability and independence from discretization density.

| Mesh Domain Level    | Total CFD Cells    | Total FEM Elements | Drag Coefficient ( $C_d$ ) | Peak Von Mises Stress ( $\sigma_{\max}$ ) | $\Delta C_d$ % |
|----------------------|--------------------|--------------------|----------------------------|-------------------------------------------|----------------|
| Coarse Grid          | $1.20 \times 10^6$ | 185,000            | 0.442                      | 232.5 MPa                                 | -              |
| Mesh Domain Level    | Total CFD Cells    | Total FEM Elements | Drag Coefficient ( $C_d$ ) | Peak Von Mises Stress ( $\sigma_{\max}$ ) | $\Delta C_d$ % |
|                      | $10^6$             |                    |                            |                                           |                |
| Medium Grid          | $2.85 \times 10^6$ | 410,000            | 0.418                      | 216.8 MPa                                 | 5.42%          |
| Fine Grid (Selected) | $4.85 \times 10^6$ | 742,150            | 0.409                      | 210.0 MPa                                 | 2.15%          |

The minimal variation observed between the Medium and Fine grid levels ( $\Delta C_d < 2.5\%$ ) confirmed that the Fine grid yielded asymptotic mesh independence. [Insert Figure 2: Mesh]

## 12. FEM Results

The linear static structural analysis yielded full 3D spatial field contours of structural deformation and Von Mises equivalent stress across the rocket casing.

### Total Structural Deformation

The maximum total deformation field recorded under combined maximum aerodynamic pressure mapping and  $15.0 \text{ kN}$  motor axial thrust was  $1.85 \text{ mm}$ .

- **Deformation Distribution:** The peak deformation was located at the tip corners of the rear stabilizing fins. This phenomenon is driven by bending moments caused by high aerodynamic drag forces acting along the fin span.
- **Airframe Shell Deflection:** The main cylindrical body tube exhibited minimal radial deflection ( $< 0.12 \text{ mm}$ ), confirming structural stiffness along the primary motor shell.

### Von Mises Stress Field Analysis

The Von Mises stress contours demonstrated a clear non-uniform stress distribution across structural members:

- **Peak Stress Location:** The global maximum Von Mises stress ( $\sigma_{\text{vm, max}} = 210.0 \text{ MPa}$ ) occurred at the trailing-edge root attachment fillet connecting the rear stabilizing fins to the cylindrical casing.

- **Secondary Peak Stress:** A secondary stress peak ( $\sigma_{vm} = 142.5 \text{ MPa}$ ) was located at the structural shoulder geometry transitioning between the tangent ogive nose cone and the cylindrical payload bay casing.
- **Nominal Shell Stress:** Stress levels across the primary cylindrical casing wall averaged between  $45.0 \text{ MPa}$  and  $65.0 \text{ MPa}$ , remaining well below critical yield thresholds.

Stress Distribution Profile:

Nose Cone Apex:  $\sim 25.0 \text{ MPa}$  (Low)

Nose-Body Shoulder:  $\sim 142.5 \text{ MPa}$  (Moderate Peak)

Main Cylinder Shell:  $\sim 55.0 \text{ MPa}$  (Nominal)

Fin Root Attachment Fillet:  $\sim 210.0 \text{ MPa}$  (Global Maximum)

[Insert Figure 3: Von Mises Stress]

## 13. CFD Results

The steady-state 3D density-based CFD simulation successfully resolved complex supersonic fluid mechanics phenomena across the vehicle surface and domain farfield.

### Aerodynamic Pressure Field Contours

- **Stagnation Point Pressure:** Maximum static pressure ( $p_0 = 188.50 \text{ kPa}$ ) was recorded at the apex tip of the tangent ogive nose cone, where the relative flow velocity decelerates to zero.
- **Expansion Regions:** Flow accelerates rapidly as it traverses the curved contour of the ogive nose cone. Upon reaching the nose-body shoulder junction, the flow encounters a geometric discontinuity, creating an expansion fan. Here, static pressure drops to a minimum local value of  $14.20 \text{ kPa}$ .
- **Fin Pressure Field:** High static pressure ( $125.0 \text{ kPa}$ ) develops along the swept leading edges of the four stabilizing fins due to localized shock wave detachment, generating significant structural normal forces.

### Velocity Field & Mach Number Contours

- **Bow Shock Formation:** A strong detached oblique shock wave originates at the nose apex. Flow transitions across this shock from supersonic inlet conditions ( $M_\infty = 2.0$ ) to a post-shock subsonic speed near the nose tip, before accelerating back to  $M = 1.82$  along the body cylinder.
- **Fin Shock-Boundary Layer Interaction:** Oblique shock waves generated by fin leading edges interact with the boundary layer developing along the main airframe, forming localized flow separation bubbles along the fin-body junction.
- **Base Wake Formation:** Behind the flat base wall of the rocket motor casing, flow detaches completely, creating a low-pressure recirculation zone. The Mach number inside this wake region drops below  $0.15$ , accompanied by a significant base drag penalty.

Compressible Flow Regimes (Mach 2.0 Flight):

\ Oblique Shock Wave

\ \_\_\_\_\_

====> M=2.0 (A) Apex Stagnation (M=0) -> (B) Expansion Shoulder (M=2.15)  
 / \_\_\_\_\_  
 / Oblique Shock Wave

## Aerodynamic Drag Component Breakdown

The total drag force calculated by integrating surface pressure vectors and viscous shear stress tensors across the complete airframe was  $F_{\text{drag}} = 1145.8\text{ N}$ .

$\text{Total Drag } (C_d) = \text{Pressure Drag } (C_{d,p}) + \text{Viscous Drag } (C_{d,v})$

| Drag Contribution Source                                 | Force Magnitude (N) | Drag Coefficient ( $C_d$ ) | % Contribution |
|----------------------------------------------------------|---------------------|----------------------------|----------------|
| <b>Wave / Pressure Drag (<math>C_{d,p}</math>)</b>       | 882.3               | 0.315                      | 77.0%          |
| <b>Viscous Skin Friction Drag (<math>C_{d,v}</math>)</b> | 263.5               | 0.094                      | 23.0%          |
| <b>Total Aerodynamic Drag (<math>C_d</math>)</b>         | <b>1145.8</b>       | <b>0.409</b>               | <b>100.0%</b>  |

## Static Aerodynamic Stability

Aerodynamic static stability requires that the Center of Pressure (CP) reside aft of the vehicle Center of Gravity (CG).

- **Center of Gravity (CG):** Located at  $x_{\text{CG}} = 1.42\text{ m}$  measured from the nose tip.
- **Center of Pressure (CP):** Calculated by integrating aerodynamic normal force moments:  $x_{\text{CP}} = 1.78\text{ m}$ .
- **Static Margin (SM):**  $SM = \frac{x_{\text{CP}} - x_{\text{CG}}}{D} = \frac{1.78 - 1.42}{0.20} = 1.80\text{ calibers}$  A static margin of  $1.80\text{ calibers}$  falls within the standard aerospace stability window ( $1.5 - 2.0\text{ calibers}$ ), confirming inherent passive aerodynamic flight stability without requiring active thrust vector control.

[Insert Figure 4: Velocity Contours]

[Insert Figure 5: Pressure Contours]

# 14. Discussion

The integrated fluid-structure interaction analysis highlights the direct correlation between external flow physics and structural airframe stresses. High-speed compressible flow phenomena—specifically shock formation, boundary layer separation, and flow expansion—dictate the surface pressure maps that drive structural stress concentrations. The CFD solution demonstrates that wave drag accounts for approximately 77% of total aerodynamic resistance at Mach 2.0. The primary source of wave drag is the bow shock originating at the nose apex, supplemented by secondary oblique shocks generated along the fin leading edges. The flow acceleration observed across the tangent ogive shoulder reduces local surface static pressure to  $14.20\text{ kPa}$ , creating a net pressure differential across the nose-body joint shell.

When mapped into the structural FEM solver, these localized pressure differentials create bending moments that concentrate stresses at geometric discontinuities. The maximum Von Mises stress ( $\sigma_{vm, max} = 210.0 \text{ MPa}$ ) occurs precisely at the fin-body attachment fillets. This stress spike results from the combined effects of spanwise aerodynamic lift/drag bending moments acting on the fins and axial compressive stresses induced by engine thrust ( $15.0 \text{ kN}$ ).

The convergence behavior observed across both solvers confirms grid independence. The boundary layer inflation scheme ( $y^+ < 1$ ) ensured accurate prediction of viscous skin friction drag ( $C_{d, v} = 0.094$ ) and prevented premature boundary layer separation estimates along the airframe body.

## 15. Engineering Interpretation

From an aerospace system design perspective, these numerical findings confirm that the proposed sounding rocket airframe is structurally viable and aerodynamically stable within the tested flight envelope.

### Structural Weight Optimization Opportunities

The structural analysis reveals that average stress levels across the main cylindrical casing wall range between  $45.0 \text{ MPa}$  and  $65.0 \text{ MPa}$ . Given that Aluminum 7075-T6 exhibits a yield strength of  $503.0 \text{ MPa}$ , the cylindrical casing shell is significantly under-stressed.

This structural margin suggests an opportunity for mass optimization:

- **Shell Thickness Reduction:** Reducing the cylindrical casing wall thickness from  $3.50 \text{ mm}$  to  $2.20 \text{ mm}$  would decrease airframe dry mass while maintaining adequate structural safety margins.
- **Fillet Radius Optimization:** Because peak stresses are concentrated at the fin-body attachment joints, introducing a generous  $5.0 \text{ mm}$  parabolic blend fillet radius at the fin root would smooth the load transfer path, reducing local stress peaks by up to 25%.

### Aeroelastic and Thermal Considerations

While static structural integrity is verified, supersonic flight introduces potential aeroelastic risks:

- **Fin Flutter:** The high aerodynamic dynamic pressure acting along the fin span induces bending forces that produce peak tip deflections of  $1.85 \text{ mm}$ . A dedicated dynamic aeroelastic modal analysis should be performed to ensure fin bending-torsion flutter frequencies do not overlap with natural airframe structural frequencies.
- **Aerodynamic Heating:** At Mach 2.0, stagnation temperatures reach approximately  $400.0 \text{ K}$  ( $127^\circ\text{C}$ ). Thermal gradients across the nose tip will introduce localized thermal stresses that must be integrated into future thermo-structural analyses.

## 16. Factor of Safety Calculation

The structural Factor of Safety (FOS) evaluates the margin between maximum working stresses and material strength limits. It is defined mathematically as:  $FOS =$

$\frac{\sigma_{yield}}{\sigma_{vm, max}}$  where:

- $\sigma_{\text{yield}}$  is the yield strength of the selected structural material (Al 7075-T6) = 503.0 MPa.
- $\sigma_{\text{vm, max}}$  is the maximum equivalent Von Mises stress obtained from the FEM simulation = 210.0 MPa.

## Step-by-Step Mathematical Calculation

$$\text{FOS} = \frac{503.0 \text{ MPa}}{210.0 \text{ MPa}} \text{ FOS} = 2.395238 \approx 2.40$$

## Ultimate Structural Margin Evaluation

In aerospace structural design standards (e.g., NASA-STD-5001 or MIL-HDBK-5), the required minimum yield Factor of Safety under ultimate flight loads is typically set between 1.25 and 1.50.

$$\text{Margin of Safety (MS)} = \text{FOS} - 1 = 2.395 - 1 = +1.395$$

Since the calculated yield Factor of Safety (FOS = 2.40) substantially exceeds the required aerospace minimum (FOS<sub>req</sub> = 1.50), and the Margin of Safety is positive (MS = +1.40), the structural design provides a robust buffer against material failure during supersonic flight.

## 17. Conclusion

This integrated numerical study provides a rigorous aerodynamic and structural performance evaluation of a supersonic sounding rocket airframe.

1. **Aerodynamic Performance:** High-resolution compressible CFD simulations at Mach 2.0 mapped key supersonic flow structures, including a detached bow shock, shoulder expansion fans, fin shock interactions, and base wake recirculation zones. The total aerodynamic drag coefficient was established as  $C_d = 0.409$ , with wave drag accounting for 77% of total vehicle resistance.
2. **Aerodynamic Stability:** Static stability calculations established a static margin of 1.80 calibers ( $x_{\text{CP}} = 1.78 \text{ m}$  vs.  $x_{\text{CG}} = 1.42 \text{ m}$ ), confirming inherent aerodynamic stability without requiring active flight control systems.
3. **Structural Integrity:** Structural FEM simulations under mapped maximum dynamic pressure fields and 15.0 kN axial thrust identified peak Von Mises stresses localized at the trailing fin-root fillets ( $\sigma_{\text{vm, max}} = 210.0 \text{ MPa}$ ).
4. **Safety Margin Verification:** The resulting yield Factor of Safety was calculated as FOS = 2.40 against the yield limit of Aluminum Alloy 7075-T6 (503.0 MPa). This confirms that the airframe design meets aerospace safety criteria under primary supersonic flight loads.

## 18. Future Scope

- **Coupled Transient Fluid-Structure Interaction (FSI):** Implement 2-way dynamic FSI modeling to evaluate aeroelastic flutter, fin oscillation modes, and transient structural responses during atmospheric transits.
- **Aerothermal Heating Coupling:** Integrate high-speed compressible thermal CFD with thermal-stress FEM to evaluate material property degradation and thermal stress distribution across structural members.

- **Trajectory-Based Transient CFD:** Extend steady-state flight analyses to full-trajectory transient simulations covering trans-atmospheric ascent from Mach 0.8 (transonic drag rise) through Mach 4.0+.
- **Composite Airframe Optimization:** Investigate weight savings by replacing metallic casing components with carbon-fiber reinforced polymer (CFRP) composite laminates, utilizing Tsai-Wu failure criteria for structural evaluation.
- **Experimental Wind Tunnel Validation:** Validate numerical pressure maps, shock locations, and drag coefficients using scale-model wind tunnel testing and high-speed Schlieren imaging.

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